Surface EMG Multichannel Measurements Using Active, Dry Branched Electrodes

Federico N. Guerrero and Enrique Spinelli

LEICI—Departamento de Electrotecnia, Universidad Nacional de La Plata, La Plata, Argentina
and Consejo de Investigaciones Cientficas y Tcnicas (CONICET), Argentina
{spinelli,federico.guerrero}@ing.unlp.edu.ar

Abstract— **A surface electromiography (sEMG) multichannel aquisition system was implemented using dry electrodes. Branched electrodes were used to obtain spatially selective EMG measurements. Active electrodes coupled with a high dynamic range converter were used to achieve good electromagnetic interference rejection and measurement robustness. Functioning parameters were measured. The device has an equivalent input referred noise less than 1 μV_{rms} with a voltage gain of 4 , allowing for a $\pm$550 mV electrode offset. The device bandwidth ranges from DC to 500 Hz with a 2000 sps sampling rate. The CMRR is 90 dB further improved by an independent DRL circuit with gain above 30 dB at 50 Hz. Multichannel EMG measurements were successfully conducted and example results are shown.**

Keywords— **surface electromiography; branched electrodes; multichannel measurements; sigma-delta converters.**

I Introduction

Surface electromyography (sEMG) measurements are a non-invasive means to obtain information about muscle activity, delivering signals that can be of use for medical diagnosis, prosthetics and rehabilitation devices. Improving the patient comfort and simplifying the use of the measurement system are desirable goals because they encourage a more frequent use of the device, and reduce the training required of the user. For this purpose, dry electrodes are an attractive solution compared with traditional wet-gel electrodes. They do not require any electrolyte other than the natural moist of the skin so they can be easily put into place, and they have no disposable parts contributing to their ease of use and simple maintenance.

The most common electrode configuration for sEMG acquisition is a simple differential pair. Many other spatial configurations have been proposed with the aim of crosstalk reduction; the double-differential variety is found in many devices. In this work, branched electrodes (as defined in [1]) were used because they achieve a good selectivity comparable with that of double-differential electrodes [2], but with a simpler implementation. The branched electrode basic design can be seen in Fig. 1. It is recommended in [3] to build the electrodes with rectangular 10 mm x 1 mm rods with a separation between 20 and 10 mm. A 10 mm inter-electrode distance is found to produce good crosstalk reduction in [4].

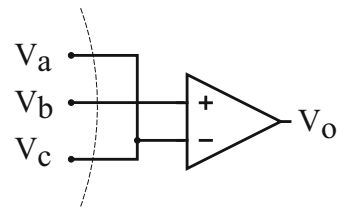

Fig. 1: Branched electrode configuration.

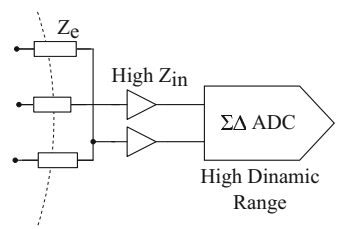

Fig. 2: Proposed implementation to manage the large impedance Z_e, voltage offset, and baseline drift of dry electrodes: high input impedance (Z_{in}) active electrodes mitigate EMI effects and a high dynamic range A/D converter allows for low gain stages.

As with any electrode, the interface between the metal contacts and the skin produces an electronic connection with non-zero impedance. In the case of dry electrodes placed on the skin without preparation, this impedance is of relatively high magnitude and variability, making them susceptible to electromagnetic interference (EMI). This problem can be overcome through electronic design strategies. Interference mechanisms are well explained in [5]: common mode signals can interfere through mode transformations caused by electrode impedances (through the "potential divider effect") and component mismatch. The baseline of signals obtained with dry electrodes takes more time to stabilize. In this work this is overcome by the use of active electrodes [6]. Large DC offsets and high amplitude artifacts can be tolerated if the amplifier input has sufficient dynamic range (DR). A DC coupled approach using Sigma Delta converters can reach a 120 dB DR, enough for EMG measurements accepting a $\pm 5V$ excursion of the baseline signal. The diagram in figure 2 summarizes the proposed approach.

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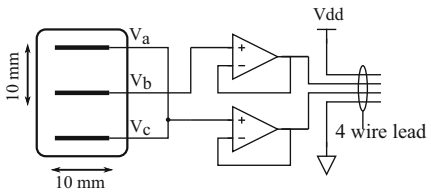

Fig. 3: Active electrodes layout.

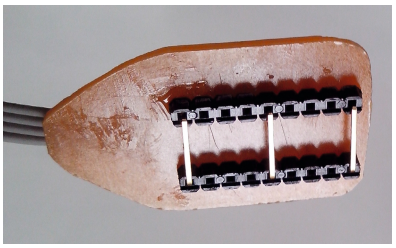

Fig. 4: The implemented active electrode. Three parallel gold-plated rod electrodes with a 10 mm inter-electrode distance were fixed to a PCB board that also holds the operational amplifiers. They provide a buffered differential output. Additional power wires are needed.

II Materials and Methods

A System implementation

The electrodes were manufactured with gold-plated standard pin connectors, as they have the exact dimensions required and are robust, with a durable surface, and commonly available. Three of these connectors were soldered parallel to each other on a printed circuit board (PCB) to implement each branched electrode. Caution was taken to make sure that the solder would not come into contact with the skin, because it was observed in previous experiences that the "solder-skin interface" was highly noisy, hampering measurements. The electrode layout can be seen in figure 3. To accomplish the requirement of high input impedance and maintain a low noise floor, operational amplifiers (op-ams) with unity gain feedback configuration were used as buffers on the electrode itself. This has the additional advantage of driving to ground interference currents capacitively coupled to the electrode wires running from the main circuit [7], which would otherwise result in significant interference.

Thus, each electrode presents a fully differential channel that extends to the converter. Bacause mismatch between components on each rail of the channel degrade the CMRR, it is impractical to strive for a ratio above 90 dB when using discrete off-the-shelve components. High quality measurements however require a CMRR greater than 100 dB [8], so a DRL circuit was used to boost the CMRR. Traditional DRL circuits average the input from every measurement channel to calculate the common mode signal

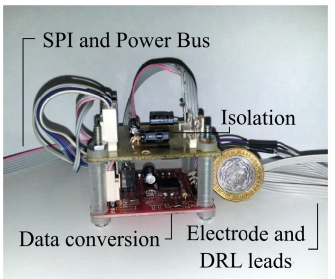

Fig. 5: Data conversion and isolation module, composed of two 5cm x 5cm boards.

to be canceled. Since dry electrodes can be accidentally detached more easily than the wet-gel kind, this is impractical as failure of one electrode would compromise the entire set of measurements. Hence, an independent DRL circuit was used, with its own measurement and feedback channels. Stability was ensured by dominant pole compensation, yielding a gain, and therefore a CMRR improvement, of nearly 30 dB at 50 Hz.

The acquisition system used in this work was based on the design presented in [9]. The modules of that device were adapted for improved portability and modified to hold a ADS1299 analog front-end from Texas Instruments. This front-end provides adjustable gain and 24-bit Sigma-Delta converters for each of its 8 channels. Two 5×5 cm circuit boards were implemented for the front-end and the isolation circuitry respectively, as can be seen in figure 5. A voltage reference is also available for the DRL. Digitized data is transmitted through a SPI bus. Isolation was implemented with two ADUM6401 ICs that provide both an isolated SPI data bus and an isolated power bus. One of them provides the $3.3V$ digital supply that the ADS1299 requires and the other one its $5V$ analog supply. A microcontroller receives the data as host of the SPI bus and relays it to a PC through a USB real-time connection to the PC. A C# application was created with a simple waveform visualization screen and a configuration panel to set the channel gain and sampling frequency, together with a logging system that creates a document with details of the acquisition sessions. Real-time visualization was included in order to have feedback over the tests (e.g., check for electrode detachment, biofeedback of muscle activation). Recorded signals were processed with Matlab.

B Measurement methods

The system was characterized by measuring parameters of interest: noise, bandwidth, and CMRR.

Noise was measured short-circuiting the branched